GOOGLE AI BUILDER CUP 2026 · FUTURE OF WORK & ENTERPRISE PRODUCTIVITY

KnowBridge
Connecting knowledge to confident decisions

Singapore-first evidence catalog for Google Workspace SMEs.

Live demo: knowbridge-208529907910.asia-southeast1.run.app · All demo data is fictional synthetic data.

THE PROBLEM

Project commitments live in scattered files — and the files disagree

- Customer targets sit in one Doc, QA forecasts in another, security sign-off in a third.
- Nothing reconciles **baseline vs forecast vs commitment vs approved decision**.
- The cost of a wrong commitment: promising a date the evidence does not support.

> *“Can we commit to Project Harbour launching on 2026-11-01?”*

CUSTOMER & MARKET

A testable hypothesis, not a market claim

- Target: Singapore services SMEs already running on Google Workspace.
- Workflow fit: **connect** → **investigate** → **reconcile** → **decide**, inside tools teams already use.
- We state this as a hypothesis to validate; no customer adoption or productivity gain is claimed.

PRODUCT

The 4-step evidence workflow

1 · Connect

Bounded selected-file import via Google Picker + OAuth. Registered fictional files only.

2 · Investigate

Grounded Ask with literal, click-through citations. Absent evidence → “insufficient evidence”.

3 · Reconcile

Temporal & conflict analysis: a later forecast never erases an earlier customer expectation.

4 · Decide

Decision Brief with options, risks, unknowns — for human review. Never an autonomous decision.

DEMO · ACTUAL DEPLOYED ANSWER

The Harbour conflict, kept typed

- Client target **2026-11-01** (HBR-DOC-02, recorded 2026-10-07)
- Latest QA forecast **2026-11-11** (HBR-DOC-04, recorded 2026-10-10) — a forecast, not a revised commitment
- Security approval PENDING (HBR-DOC-06) — pending is not approved
- No revised commitment authorized (HBR-DOC-09) — recommendations stay conditional

Every claim carries a literal excerpt and a source link; fabricated citations are rejected server-side.

TECHNICAL ARCHITECTURE

Cloud Run · Gemini · Firestore — fail-closed grounding

- Cloud Run (asia-southeast1) · Gemini **gemini-3.8-flash** (global) · **gemini-embedding-001** 768-d vectors (us-central1) · Firestore
- Gates before any answer ships: tenant-scoped retrieval → literal citation verification → decision coverage → semantic review → fail closed
- Security basics: non-root container, Secret Manager, revocation-checked Firebase tokens on mutations, CSP, synthetic-tenant pinning

Honest disclosure: model processing is not Singapore-only. Real-data import is prohibited.

EVALUATION · ACTUAL MEASURED RESULTS

Honest numbers, including failures

168 / 24 / 1,904
fictional Docs / Sheets / rows — seed verifier PASS

256 / 256
unit & integration tests PASS · lint · typecheck · build PASS

2 / 12
both final deployed benchmark runs (before/after the reliability fix): every failure was a provider-side service error (HTTP 503), never an evidence-quality failure

9 / 10 / 11 of 12
earlier complete runs — totals are never aggregated across runs

Full-suite reliability under intermittent provider-side model failures remains an open issue and is reported as such.

DIFFERENTIATION & ROADMAP

Explainable, temporally aware decisions

- Typed evidence semantics: baseline / forecast / commitment / approved decision are never conflated.
- Server-side integrity: every excerpt must literally exist in its source, or the answer is rejected.
- Abstention is a feature: missing evidence is reported, never invented.
- Roadmap: deeper Workspace integration in scope · Gmail/Chat/Calendar and SSO post-hackathon.

KnowBridge — Know what your sources actually say.